

R Packages alpha

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Apply a 'regularized log' transformation

Description

This function uses Tikhonov/ridge regularization to transform the data to the log₂ scale in a way which minimizes differences between samples for rows with small counts. The rlog transformation produces a similar variance stabilizing effect as `varianceStabilizingTransformation`, though `rlogTransformation` is more robust in the case when the size factors vary widely. The transformation is useful when checking for outliers or as input for machine learning techniques such as clustering or linear discriminant analysis. Note: `rlog` is equivalent to `rlogTransformation`, which take as input a `DESeqDataSet` and return a `SummarizedExperiment` object.

Usage

```
1 rlog(object, blind = TRUE, fast = FALSE, intercept, betaPriorVar, B)
2
3 rlogTransformation(object, blind = TRUE, fast = FALSE, intercept,
4   betaPriorVar, B)
5
6 rlogData(object, intercept, betaPriorVar)
7
8 rlogDataFast(object, intercept, betaPriorVar, B)
```

Arguments

`object`

a `DESeqDataSet`

`blind`

logical, whether to blind the transformation to the experimental design. `blind=TRUE` should be used for comparing samples in an manner unbiased by prior information on samples, for example to perform sample QA (quality assurance). `blind=FALSE` should be used for transforming data for downstream analysis, where the full use of the design information should be made. If many of genes have large differences in counts due to the experimental design, it is important to set `blind=FALSE` for downstream analysis.

`fast`

if `TRUE`, an approximation to the rlog transformation is made, wherein an optimal amount of

shrinkage of sample-wise log fold changes is calculated for genes divided into 12 batches based on mean counts. then shrinkage is performed by interpolating between these optimal values. optimal is defined by maximizing the same posterior as in the standard rlog transformation.

intercept

by default, this is not provided and calculated automatically. if provided, this should be a vector as long as the number of rows of object, which is log2 of the mean normalized counts from a previous dataset. this will enforce the intercept for the GLM, allowing for a "frozen" rlog transformation based on a previous dataset.

betaPriorVar

a single value, the variance of the prior on the sample betas, which if missing is estimated from the data

B

for the fast method only. by default, this is not provided and calculated automatically. if provided, this should be a vector as long as the number of rows of object. this is the amount of shrinkage from 0 to 1 for each row, and takes precedence over internal calculation of B using betaPriorVar.

Details

Note that neither rlog transformation nor the VST are used by the differential expression estimation in DESeq, which always occurs on the raw count data, through generalized linear modeling which incorporates knowledge of the variance-mean dependence. The rlog transformation and VST are offered as separate functionality which can be used for visualization, clustering or other machine learning tasks. See the transformation section of the vignette for more details.

The 'regularization' referred to here corresponds to the maximum a posteriori solution to the GLM with a prior on the coefficients for each sample. The fitted dispersions are used rather than the MAP dispersions (so similar to the varianceStabilizingTransformation) as the blind dispersion estimation would otherwise shrink large, true log fold changes. The prior variance is calculated as follows: a matrix is constructed of the logarithm of the counts plus a pseudocount of 0.5, the log of the row means is then subtracted, leaving an estimate of the log fold changes per sample over the fitted value using only an intercept. The prior variance is then calculated as with nbinomWaldTest, by matching the upper quantiles of the observed log fold change estimates with an upper quantile of the normal distribution. A second and final GLM fit is then performed using this prior. It is also possible to supply the variance of the prior. See the vignette for an example of the use and a comparison with varianceStabilizingTransformation.

The transformed values, $rlog(K)$, are equal to $rlog(K_{ij}) = \log_2(q_{ij}) = x_{.j} \cdot \beta_i$, with formula terms defined in DESeq.

The parameters of the rlog transformation from a previous dataset can be "frozen" and reapplied to new samples. See the "Data quality assessment" section of the vignette for strategies to see if new samples are sufficiently similar to previous datasets. The "freezing" is accomplished by saving the dispersion function, beta prior variance and the intercept from a previous dataset, and running rlogTransformation with 'blind' set to FALSE (see example below).

Value

`rlog`, equivalent to `rlogTransformation`, returns a `SummarizedExperiment`. The matrix of transformed values are accessed by `assay(rld)`. For `rlogData`, a matrix of the same dimension as the count data, containing the transformed values. To avoid returning matrices with NA values where there were zeros for all rows of the unnormalized counts, `rlogTransformation` returns instead all zeros (essentially adding a pseudocount of one, only to those rows in which all samples have zero).

See Also

`plotPCA`, `varianceStabilizingTransformation`

Examples

```
1 dds <- makeExampleDESeqDataSet(m=6,betaSD=1)
2 rld <- rlog(dds, blind=TRUE)
3 dists <- dist(t(assay(rld)))
4 plot(hclust(dists))
5
6 # run the rlog transformation on one dataset
7 design(dds) <- ~ 1
8 dds <- estimateSizeFactors(dds)
9 dds <- estimateDispersions(dds)
10 rld <- rlog(dds, blind=FALSE)
11
12 # apply the parameters to a new sample
13
14 ddsNew <- makeExampleDESeqDataSet(m=1)
15 mcols(ddsNew)$dispFit <- mcols(dds)$dispFit
16 betaPriorVar <- attr(rld,"betaPriorVar")
17 intercept <- mcols(rld)$rlogIntercept
18 rldNew <- rlog(ddsNew, blind=FALSE,
19               intercept=intercept,
20               betaPriorVar=betaPriorVar)
```

- DESeq: Differential expression analysis based on the negative...
- DESeqDataSet: DESeqDataSet object and constructors
- DESeqResults: DESeqResults object and constructor
- coef: Extract a matrix of model coefficients/standard errors
- collapseReplicates: Collapse replicates in a SummarizedExperiment or DESeqDataSet
- counts: Accessors for the 'counts' slot of a DESeqDataSet object.
- design: Accessors for the 'design' slot of a DESeqDataSet object.
- dispersionFunction: Accessors for the 'dispersionFunction' slot of a DESeqDataSet...
- dispersions: Accessor functions for the dispersion estimates in a...
- estimateDispersions: Estimate the dispersions for a DESeqDataSet
- estimateDispersionsGeneEst: Low-level functions to fit dispersion estimates
- estimateSizeFactors: Estimate the size factors for a DESeqDataSet
- estimateSizeFactorsForMatrix: Low-level function to estimate size factors with robust...
- fnkm: FPKM: fragments per kilobase per million mapped fragments
- fpm: FPM: fragments per million mapped fragments
- makeExampleDESeqDataSet: Make a simulated DESeqDataSet
- nbinomLRT: Likelihood ratio test (chi-squared test) for GLMs
- nbinomWaldTest: Wald test for the GLM coefficients
- normalizationFactors: Accessor functions for the normalization factors in a...

- `plotDispEsts`: Plot dispersion estimates
- `plotMA`: MA-plot from base means and log fold changes
- `plotPCA`: Sample PCA plot from variance-stabilized data
- `replaceOutliers`: Replace outliers with trimmed mean
- `results`: Extract results from a DESeq analysis
- `rlog`: Apply a 'regularized log' transformation
- `show`: Show method for DESeqResults objects
- `sizeFactors`: Accessor functions for the 'sizeFactors' information in a ...
- `varianceStabilizingTransformation`: Apply a variance stabilizing transformation (VST) to the...

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